Assignment 10 Comparative Evaluation of Routing Protocols Using NS-3

Deadline: 28/9/26, 01/10/26

Consider a network consisting of multiple routers to provide communication between different departments of an organization. The network uses different routing approaches such as RIP, OSPF, and AODV. The network conditions are varied by introducing congestion and link failures. Simulate this scenario using NS3.

Perform the following:

1. **Network Topology**
    - Design a multi-router network topology in NS-3 with multiple source and destination nodes.
    - Implement RIP, OSPF, and AODV routing protocols.
    - Configure suitable IP addresses and application traffic between source and destination nodes.
2. **Normal Network Condition**
    - Generate CBR/UDP traffic between selected source-destination pairs.
    - Measure:
        - Packet Delivery Ratio (PDR)
        - End-to-End Delay
        - Routing Table Updates
3. **Congestion Condition**
    - Increase the offered traffic by introducing additional simultaneous flows.
    - Measure and compare the performance of the routing protocols.
4. **Link-Failure Condition**

    Disable one of the intermediate links during simulation.

    Observe how each routing protocol responds and establishes an alternate route.

    Measure the resulting PDR, delay, and routing-table changes.
5. **Performance Analysis**

    Compare the three protocols under normal, congested, and link-failure conditions.

Plot the following graphs:

- PDR vs. Network Condition
- End-to-End Delay vs. Network Condition
- Routing Table Updates vs. Network Condition